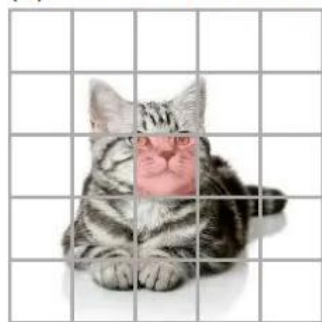
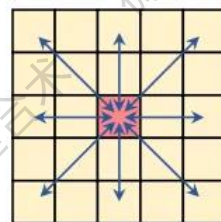


核心速览

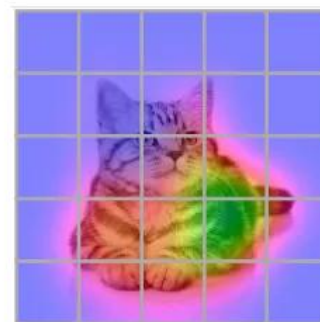
(a) Previous Methods:



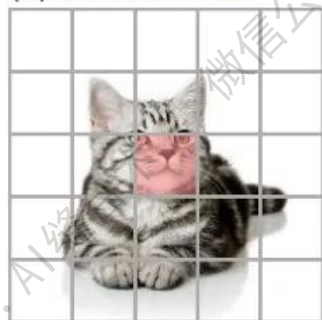
Self-Attention



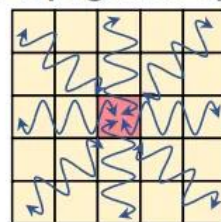
$$\text{softmax}(QK^T / \sqrt{d_k})V \sim O(N^2)$$



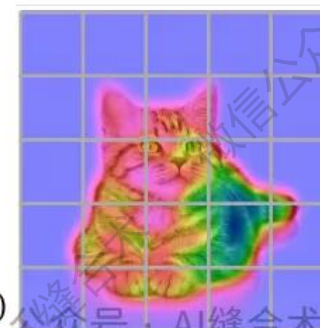
(b) WaveFormer:



Wave Propagation Operator



$$e^{-\frac{\alpha}{2}t} (A \cos(\omega_d t) + B \sin(\omega_d t)) \sim O(N \log N)$$



论文题目: WaveFormer: Frequency-Time Decoupled Vision Modeling with Wave Equation

中文题目: WaveFormer:基于波动方程的频时解耦视觉建模

论文链接: <https://arxiv.org/pdf/2601.08602>

所属单位: 北京大学, 清华大学

核心速览: 本文提出基于欠阻尼波动方程的 WaveFormer 框架, 通过频率-时间解耦传播机制, 设计 Wave Propagation Operator(WPO)实现 $O(N \log N)$ 复杂度全局交互。模型在 ImageNet-1K 分类(84.2% Top-1)、

COCO 检测 (47.9% APb) 和 ADE20K 分割 (50.5% mIoU) 任务上超越 Swin/ConvNeXt 等基准, 同时吞吐量提升 1.6 倍、FLOPs 降低 30%。

论文内容详细解读:

<https://mp.weixin.qq.com/s/hiS2HmJ0WGdYntEzrUf1Zw>

使用设备: cuda

WPO 模型结构:

```
WPO(  
  (dwconv): Conv2d(96, 96, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=96)  
  (linear): Linear(in_features=96, out_features=192, bias=True)  
  (out_norm): LayerNorm((96,), eps=1e-05, elementwise_affine=True)  
  (out_linear): Linear(in_features=96, out_features=96, bias=True)  
  (to_k): Sequential(  
    (0): Linear(in_features=96, out_features=96, bias=True)  
    (1): ReLU()  
  )  
)
```

输入张量形状 : torch.Size([1, 96, 14, 14])

输出张量形状 : torch.Size([1, 96, 14, 14])